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Genomically Smoldering Multiple Myeloma Is Not a Distinct Entity But a Collection of Monoclonal Gammopathy of Undetermined Significance or Multiple Myeloma

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ABSTRACT

PURPOSE The diagnosis of smoldering multiple myeloma (SMM) primarily relies on clinical features such as plasma cell involvement, immunoglobulin protein levels, and end-organ damage. However, as early intervention becomes a priority, the role of genomic features in differentiating risk is gaining attention.

METHODS This study analyzed next-generation sequencing data from 224 precursor condition samples with 51 patients having paired SMM and multiple myeloma (MM) and 1,779 samples from newly diagnosed MM to identify genomic features linked to progression in SMM and those with a low-risk nonprogressor precursor condition.

RESULTS Our findings from paired samples revealed no significant differences in somatic alterations and clonal structures between SMM and MM samples from the same patient. This indicates that plasma cells in progressor SMM are genomically pre-equipped with changes that define myeloma. Over 80% of driver mutations were present at both time points, and more than 66% of progressor samples showed only minor clonal changes. We further compared genomic changes between nonprogressor and progressor SMM. Nonprogressor plasma cells showed significantly lower mutational load and the absence of copy number alterations on chromosome 8. They reduced focal genomic loss compared with progressor plasma cells. A scoring system using genomic features predictively identified patients with low-risk SMM unlikely to progress, validated on 101 additional independent samples, and additive clinical value of genomic classification was further shown in combination with 20/2/20.

CONCLUSION In summary, the genomic distinctions now suggest that a proportion of SMMs with progressor phenotype are akin to MM, whereas nonprogressor SMM has monoclonal gammopathy of undetermined significance–like characteristics. The results should influence further investigation in larger studies to inform future diagnostic criteria and trial designs.

ACCOMPANYING CONTENT

Data Sharing Statement

Data Supplement

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INTRODUCTION

Monoclonal gammopathy of undetermined significance (MGUS) and smoldering multiple myeloma (SMM) are precursor stages of myeloma lacking any end-organ damage or acquisition of any of the SLiM-CRAB features[1] observed in multiple myeloma (MM).[2,3] About 1% of patients with MGUS and 10% of patients with SMM progress to MM each year in the first 5 years,[2,4-6] with risk declining after that.[4] However, some patients with high-risk SMM progress rapidly within 2 years,[7] identified by risk markers like the 20/2/20 criteria, tumor burden, and genomic alterations.[8-16] Yet, there is a broad spectrum of risk concerning patients with SMM, and improving the definition of risk, especially a proportion of patients who are not likely to progress, will allow us to design better follow-up measures and clinical trials in patients with SMM and offer more accurate prognoses.[7,17,18]

CONTEXT

Key Objective

Is smoldering multiple myeloma (SMM) a genomically distinct entity from multiple myeloma (MM), or do genomic profiles reveal that it represents a heterogeneous collection of monoclonal gammopathy of undetermined significance (MGUS)–like and MM-like diseases?

Knowledge Generated

Patients with progressor SMM already harbor MM-defining genomic alterations at diagnosis, with >80% of driver mutations present before progression and minimal clonal evolution occurring during transformation. Patients with nonprogressor SMM exhibit MGUS-like features with significantly lower mutational burden, the absence of chromosome 8 alterations, and reduced genomic instability, enabling the development of a genomic scoring system that achieved 85% sensitivity for identifying low-risk patients unlikely to progress.

Relevance *(S. Lentzsch)*

By demonstrating that SMM is not a biologically distinct entity but rather a mixture of molecularly defined MGUS or MM, this study reframes our understanding of precursor conditions. These findings lay the groundwork for using next-generation sequencing-based genomic profiling to better distinguish progressors from nonprogressors and, ultimately, guide personalized treatment decisions, potentially retiring the term "SMM" in favor of a biologically driven risk model.*

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Various genomic and clinical features have been correlated with progression time to improve the definition of risk categories.[14] Alterations in the *KRAS*, *NRAS*, and DNA-repair genes (del17p, *TP53*, *ATM*) and *MYC* have been reported as independent risk factors for SMM-to-MM progression.[11,12,15,19] Conversely, nonprogressor plasma cells (NP-PCs) arise later in life and lack chromothripsis, driver mutations, aneuploidy, and APOBEC activity.[13] However, genomic studies of NP-PC have been hampered by small sample sizes, short follow-up periods, and few paired samples.

Clinical trials have tested early intervention in SMM[20-23] using risk models like IMWG, Mayo 2008, Mayo 20/2/20, GEM-Pethema, and SLiM-CRAB.[10,16,24-26] However, these models identify a portion of patients with high-risk SMM likely to progress within 5 years.[27] While genomics helps define risk groups in MM, its use in SMM remains limited.[28-31]

Furthermore, genomic studies show that primary and subclonal events differ between newly diagnosed MM (NDMM) and SMM.[32-35] Therefore, leveraging genomics is becoming essential in SMM to differentiate patients who might already have MM-like genotypes and those who may not likely to progress in the near future. Recognizing such information can help us understand treatment decisions to avoid under- or overtreatment and to shape future studies.

Here, we evaluated whole-genome sequencing (WGS) from patients with high-risk SMM (time to progression [TTP] <5 years) at their SMM diagnosis and later at progression to MM, nonprogressor (low-risk) plasma cell precursor conditions (the minimum follow-up time is >5 years with no progression), a large cohort of NDMM to evaluate shared and added features as well as clonal evolution, and genomic changes to understand differences between low-risk SMM and progressor SMM and MM.

METHODS

All details regarding cohorts and analysis are provided in the Data Supplement (online only). In brief, the methods are explained below.

Samples

Samples from 224 patients with SMM were collected at SMM stage in separate cohorts: 64 in the training cohort, 101 in the validation cohort, and 59 additional SMM samples without follow-up information (used for mutation frequency comparison). Fifty-one patients had paired samples at MM diagnosis. Table 1 summarizes patient characteristics. One hundred and one NP-PCs and progressor plasma cells (P-PCs) from patients formed a separate validation set; four samples without known progression status were excluded. Three hundred thirty-six NDMM samples and 160 SMM samples were profiled with SNP arrays. All patients were from the University Cancer Center of Toulouse and provided informed consent after ethics committee approval. Furthermore, 1,000 NDMM-WGS samples from independent studies and 779 whole exome sequencing (WES) samples from the MMRF CoMMpass study were included. Bone marrow samples were purified using CD138+ magnetic beads, and only quality control passed samples were

TABLE 1. Clinical Profiles of Patients

Characteristic	P-PC (n = 34)	NP-PC (n = 30)	*P*	Validation 1 (n = 42)	Validation 2 (n = 59)	NDMM (n = 1,779)
Age at diagnosis, years, median (IQR)	59.53 (8.02)	56.69 (16.78)	.62	64.8 (12.35)	62 (14)	60.6 (11; missing = 16)
TTP, years, median (IQR)	1.43 (1.37)	—		1.52 (1.50)	1.46 (1.25)	NA
Median follow-up time (years)	—	7.63 (4.69)		7.37 (2.23)	8.21 (1.59)	NA
Sex, No. (%)						
Female	14 (41)	15 (50)	.21	19 (45)	34 (54)	693 (39)
Male	20 (59)	13 (43)		23 (54)	29 (46)	1,049 (59)
Missing						37 (2)
del1p32, No. (%)						
No	30 (88)	28 (93)	.68	38 (90)	54 (91)	1,540 (87)
Yes	4 (12)	2 (6)		4 (10)	5 (9)	236 (13)
gain1q21, No. (%)						
No	16 (47)	23 (77)	.021	28 (66)	42 (71)	1,153 (65)
Yes	18 (53)	7 (23)		14 (34)	17 (29)	626 (35)
del17p, No. (%)						
No	34 (100)	29 (96)	.46	40 (95)	57 (97)	1,576 (89)
Yes	0	1 (4)		2 (5)	2 (3)	203 (11)
del13, No. (%)						
No	13 (38)	17 (57)	.21	23 (55)	39 (66)	946 (53)
Yes	21 (62)	13 (43)		19 (45)	20 (34)	833 (47)
HMM, No. (%)						
No	18 (53)	13 (43)	.47	19 (45)	36 (61)	897 (51)
Yes	16 (47)	17 (57)		23 (55)	23 (39)	882 (49)
t(4;14), No. (%)						
Wild type	27 (79)	27 (90)	.31	38 (90)	52 (88)	1,548 (87)
Positive	7 (21)	3 (10)		4 (10)	7 (12)	231(13)
t(11;14), No. (%)						
Wild type	28 (82)	27 (90)	.48	34 (81)	43 (73)	1,418 (80)
Positive	6 (18)	3 (10)		8 (19)	16 (27)	361 (20)
t(14;16), No. (%)						
Wild type	34 (100)	29 (96)	.46	41(98)	56 (95)	1,703 (96)
Positive	0	1 (4)		1 (2)	3 (5)	76 (4)
MYC translocations, No. (%)						
Wild type	24 (71)	28 (93)	.029	31 (74)	37 (63)	
Positive	10 (29)	2 (7)		11 (26)	22 (37)	
20/2/20 category, No. (%)						
Low/intermediate	6 (18)	2 (6)	.039	12 (29)	20 (34)	NA
High	9 (26)	12 (40)		5 (12)	11 (19)	
Undetermined	19 (56)	16 (53)		25 (59)	28 (47)	

Abbreviations: NA, not available; NDMM, newly diagnosed MM; NP-PC, nonprogressor plasma cell; P-PC, progressor plasma cell; TTP, time to progression.

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sequenced. Tumor DNA was extracted from sorted cells; peripheral blood DNA was used as germline control.

WGS Data Processing

All DNA libraries were sequenced with ≥60× coverage for tumors. Reads were aligned to GRCh38. Mutations were called with Mutect2[36] and GATK,[37] annotated with VEP, and converted to MAF files. Manta[38] and FACETS[39] detected structural and copy number (CN) variants. MAFtools[40] generated plots and comparisons; GISTIC[41] identified recurrent CN changes.

Genomic Scar Scores and Phylogenetic Tree Construction

Allele-specific copy number alterations were used to calculate genomic scar score (GSS) using scarHRD.[42] Phylogenetic trees were generated from single nucleotide variants (SNVs), accounting for purity and ploidy. Mutation

clonality was estimated from variant allele frequency using an expectation-maximization algorithm. QuantumClone[43] clustered mutations at different stages, and ClonEvol[44] modeled clonal evolution.

SNV, Structural Variants, and Copy Number Alteration Signatures

SNV and structural variants (SV) signatures were estimated using Sigminer,[45] and copy number alteration (CNA) signatures were estimated using SigProfilerExtractor.[46] SNVs were mapped to trinucleotide contexts to create 96 mutation categories. SV signatures were classified by clustering, type (deletion, inversion, tandem duplication), and event size and then combined with interchromosomal translocations. Absolute CNs were categorized into 48 channels by heterozygosity and segment size. CN signatures were visualized using Python.

Classification Model

We developed a risk score based on five variables differing between progressor and nonprogressor SMMs: gain1q, *NRAS* mutation, chromosome 8 CN changes, GSS > 5.5, and focal deletion >9.3 Mb. Each variable scored 0 or 1 for a total score of 0-5. Scores ≤1 were low-risk; scores ≥2 were high-risk. Accuracy, sensitivity, recall, and F1 scores were calculated in the training data set with 1,000 bootstrapping to evaluate stability (Data Supplement, Fig S5). Classification sensitivity and accuracy were then tested in an independent data set. For samples with available 20/2/20 information, a combination of two scores was assessed. The Cox proportional hazards regression model was used to compare progression times between groups. The likelihood ratio test was used to assess whether the model with group information is significantly better than a null model (no predictors), and the log-rank test was used to compare groups.

Comparing Cancer Cell Fraction Distributions

We compared cancer cell fraction (CCF) distributions using a paired Kolmogorov-Smirnov (KS) test. For each patient, we extracted CCF values at both time points and computed the observed KS statistics. To estimate the null distribution of KS statistics under the assumption of no difference between time points, we pooled CCF values for each patient and performed 1,000 bootstrap resampling, drawing groups of equal size to the number of mutations and computing the KS statistics for each resample. Empirical *P* values were calculated as the proportion of null KS statistics greater than or equal to the observed value. We used the Wilcoxon signed-rank test to compare the observed KS statistics with null distributions, evaluating whether the observed differences were systematically greater than expected.

Statistical Analysis and Visualization

All *P* values were calculated using two-tailed tests unless otherwise noted. We used median (IQR) and frequency (%) to characterize the patients and events in the study. Visualizations were plotted using R.

In Figures 1B and 1C, all comparisons were made using the paired sample Wilcoxon rank-sum test. In Figure 2, CCF distributions were compared as explained above, and clonal evolution trees were formed using ClonEvol. In Figure 3, the frequency of somatic events in unrelated groups (eg, SMM *v* NDMM or P-PC *v* NP-PC) was compared using Fisher's exact test. The Wilcoxon-rank sum test was used for continuous variable comparisons in Figures 3B and 3G and 4B and 4D. In Figure 5, odds ratios, 95% CI, and *P* are calculated using Fisher's exact test for counted events (Fig 5A) and a generalized linear model was used for continuous variables (Fig 5B).

RESULTS

We have analyzed WGS data from 224 patients with SMM in three cohorts and 1,000 patients with NDMM. We have additionally combined 779 NDMM-WES data from CoMMpass and analyzed SNP array from 160 patients with SMM to 336 with patients NDMM. Fifty-one patients with SMM also had a second set of paired samples collected at the time of progression to MM. We defined risk groups over 5 years, classifying patients with no progression to MM after 5 years as nonprogressors and those progressing within 5 years of SMM diagnosis as progressors. Between the NP-PC and P-PC patients, demographics were similar (Table 1, Data Supplement, Table S1), and 44% of the patients with available data in the progressor group would be considered high-risk based on the 20/2/20[16] criteria versus <16% of the NP group.

The Majority of Patients With SMM Have the Same Driver Events at MM

We compared paired samples from 51 patients diagnosed with SMM and later progressed to MM. We first investigated the enrichment of driver mutations between the SMM and MM samples. Most driver mutations were present in both samples, with few exceptions (Fig 1A). Notably, over 80% of driver events were present at both time points, and the frequencies of mutations were similar between paired samples with comparable CCFs, suggesting that mutations maintain a consistent level of clonality from the SMM to MM.

Similarly, a majority of CNAs and translocations were detected at both time points (Fig 1A). CNAs in regions associated with MM risk, such as del1p32 and gain1q21, were found in both samples with increased frequency of gain1q at MM for five patients. Interestingly, in one SMM with subclonal gain1q, the abnormality was not detected in the MM. In addition, five other patients exhibited gain1q at the time of MM diagnosis. A subclonal del17p was the only event found exclusively in the MM diagnosis for one patient (Fig 1A, Data Supplement, Table S2).

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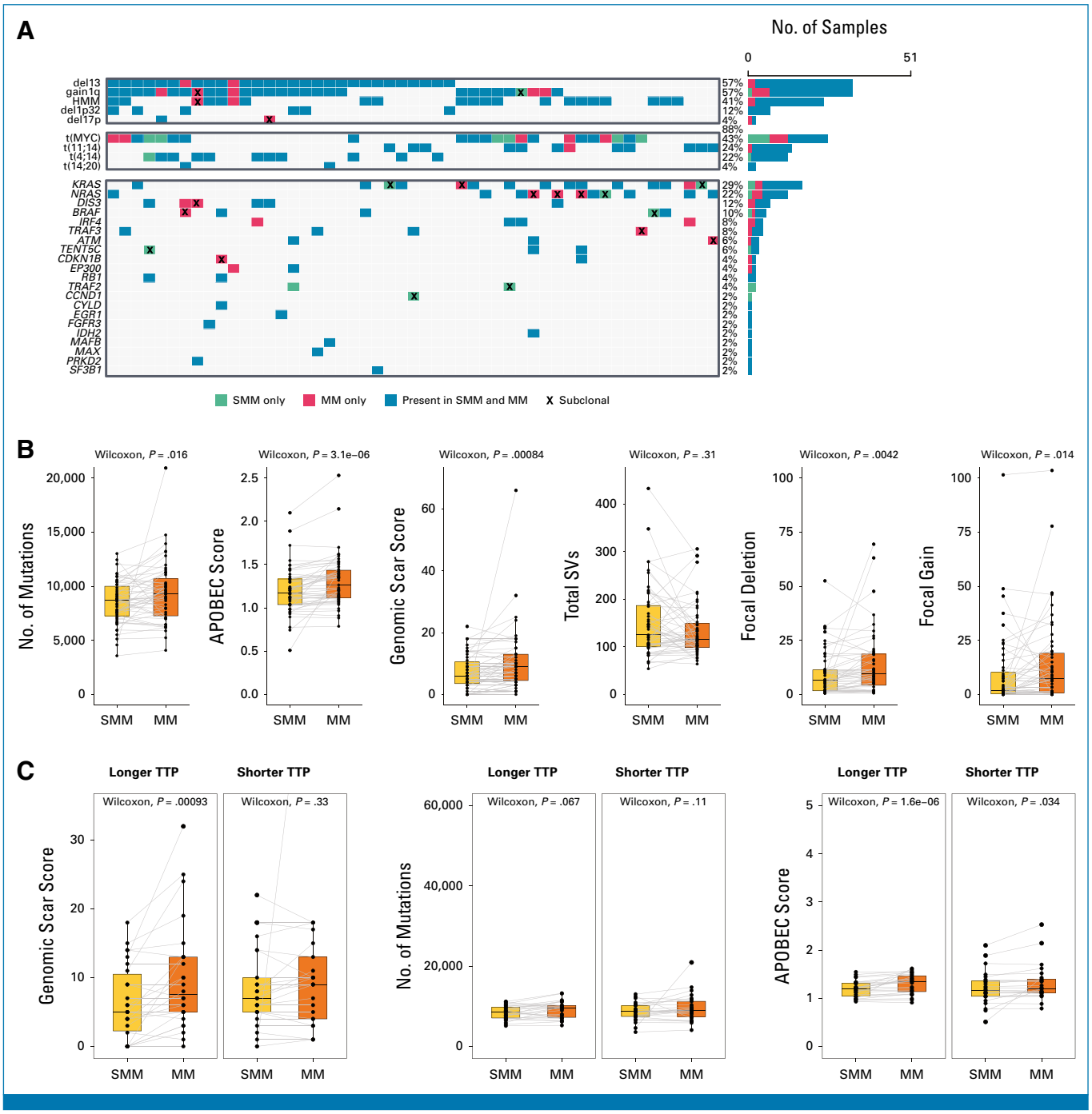

FIG 1. Comparison between SMM and MM samples collected from the same patient. (A) The somatic profiles of 51 patients, which include samples from precursor conditions and paired MM samples, are displayed according to each patient's driver mutations, CN alterations, and translocations. Each column represents a single patient. Somatic events shared between SMM and MM samples from the same patient are indicated in blue. Events unique to SMM are shown in green, whereas events specific to MM are highlighted in red boxes. The upper panel illustrates the mutational differences between each patient's MM and SMM conditions. The bar plots on the right panel depict the frequency of each somatic event, color-coded by whether they are shared (blue) or time-specific (SMM only is green, and MM only is red). Subclonal events related to CN alterations and mutations are marked with a cross (X). (B) The box plots, arranged from left to right, illustrate the differences between paired samples of SMM (in yellow) and MM (in orange) for six metrics: total mutations, APOBEC enrichment score, GSS, total number of structural variants, focal gains, and deletions. The Wilcoxon rank-sum test *P* values for the paired comparisons between the two groups are displayed at the top of each plot. GSS, number of total mutations, and APOBEC enrichment score difference between SMM (yellow) and MM (orange) pairs for a long TTP (left) and a short TTP (right) patients. Short TTP is defined as shorter than median TTP. CN, copy number; GSS, genomic scar score; MM, multiple myeloma; SMM, smoldering multiple myeloma; SV, structural variants; TTP, time to progression.

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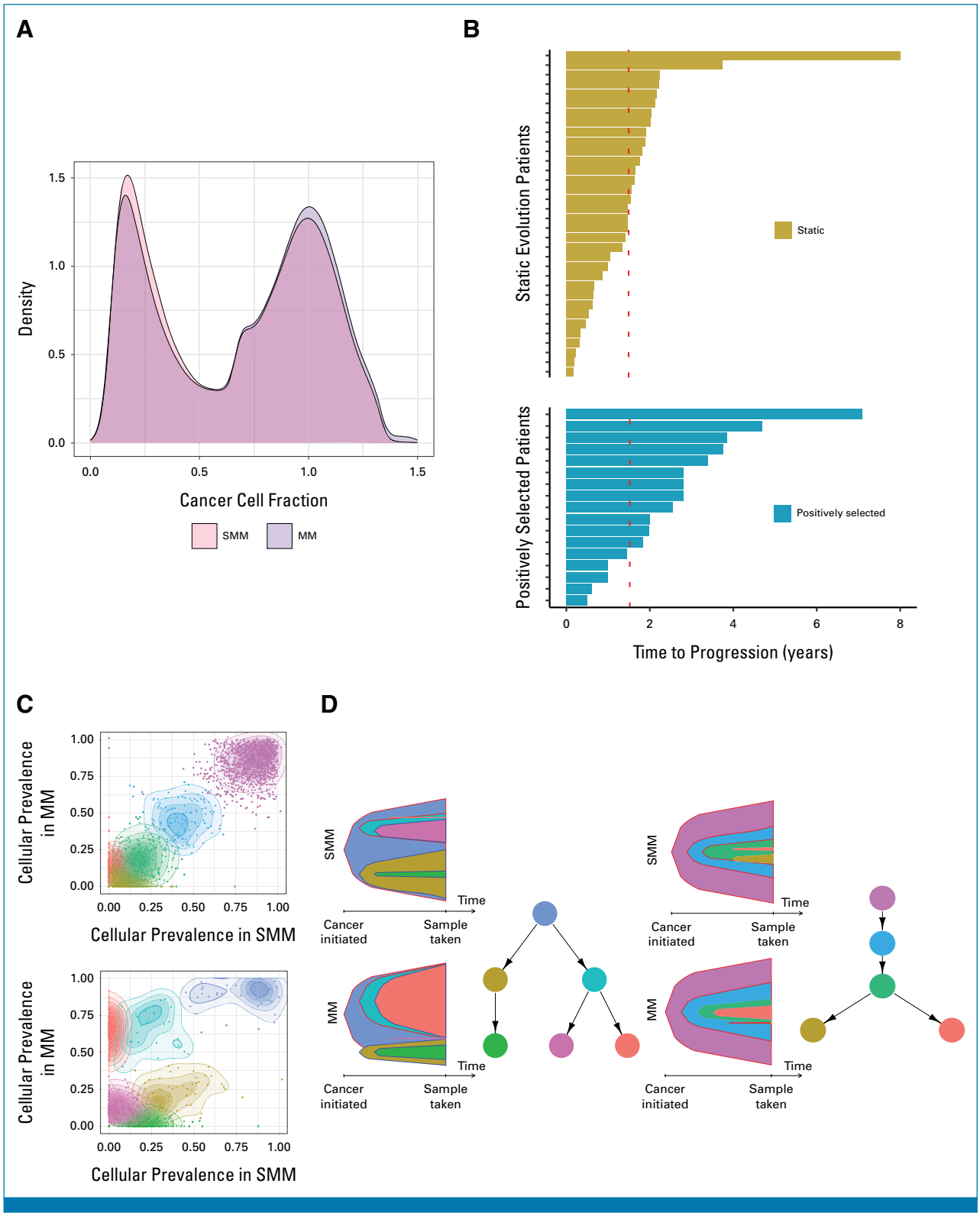

FIG 2. Evolution patterns and comparison between short- and long-term progressors. (A) Cancer cell fraction (*x*-axis) densities (*y*-axis) for all mutations at SMM (pink) and MM (purple) conditions for paired samples. (B) Genomic evolution patterns (color-coded) and TTP for SMM and MM pairs. (C) The mutation clusters displayed in two dimensions represent a fixed evolutionary pattern for one patient and clonal selection for another. Mutations are clustered and color-coded using the EM algorithm based on cancer cell fractions at the precursor condition (*x*-axis) and MM (*y*-axis). (D) Phylogenetic trees illustrate the clonal evolution patterns for the same two patients shown in (C). Each cluster's CCF is proportional on the *y*-axis, along with a fish plot and predicted trees next to the fish plots. Example on the left is from a positive selection patient, and example on the right is from a static evolution patient. CCF, cancer cell fraction; MM, multiple myeloma; SMM, smoldering multiple myeloma; TTP, time to progression.

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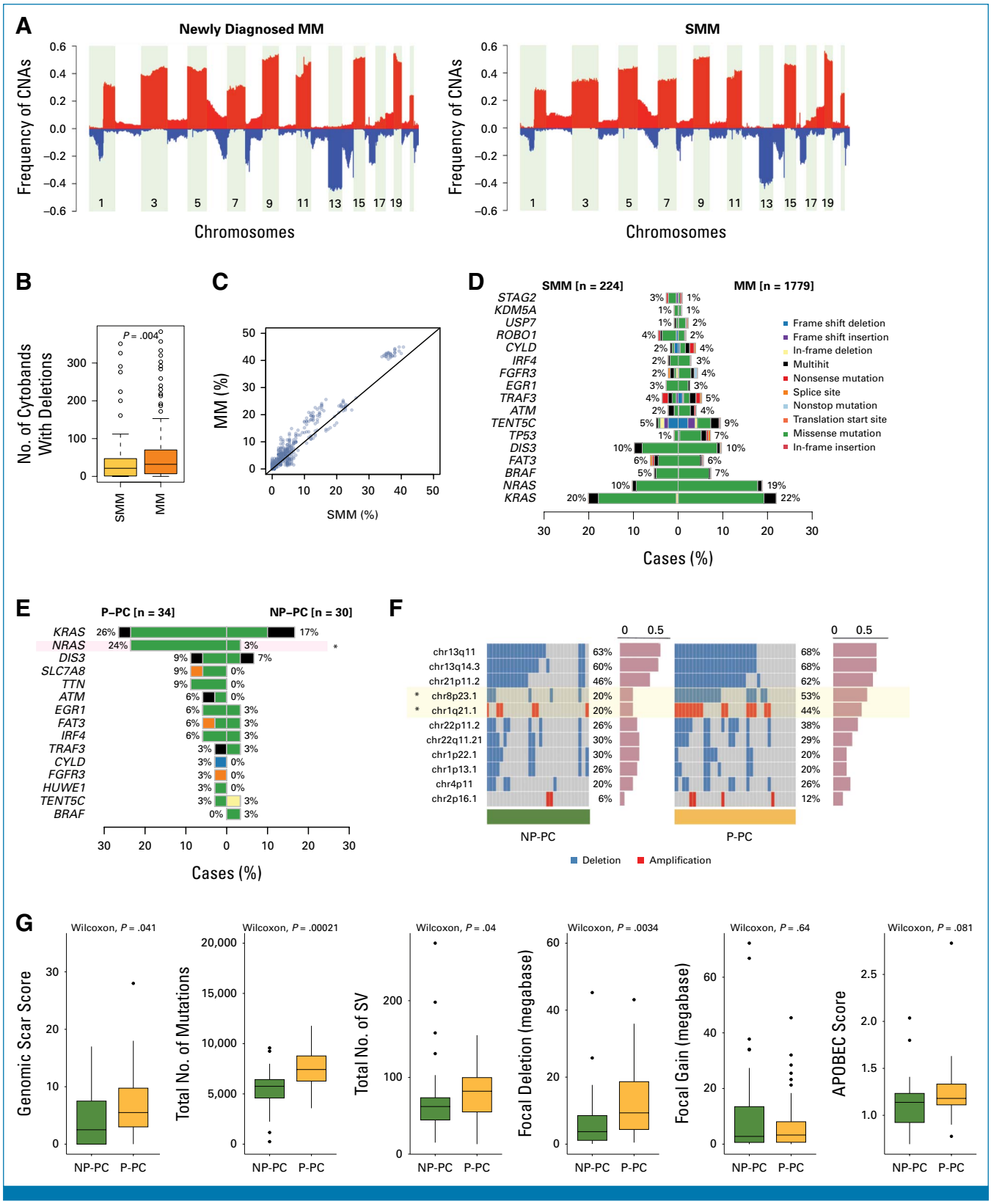

FIG 3. Genomic landscape between large-scale SMM and MM cohorts and progressor and nonprogressor samples at precursor condition. (A) The CN profiles of patients with NDMM (n = 336) and patients with SMM (n = 160) are displayed. In the profiles, red represents CN gains, whereas blue indicates deletions. The *y*-axis shows the frequency of each genomic event, and the *x*-axis spans genomic regions from chromosomes 1 to 22. (B) The number of cytobands (*y*-axis) with CN loss for SMM (yellow) and MM (orange) samples from (A) is given. The *P* value for the comparison is given at the top. (C) Scatter plot of frequency of deletions for each cytoband (blue dots) at SMM (*x*-axis) and MM (*y*-axis). The diagonal line shows where $x = y$. (D) Driver mutation (*y*-axis) frequencies (*x*-axis) for 224 precursors (left side) and 1,779 samples from NDMM (right side). Mutations are color-coded by variant type, as shown in the figure legend. (continued on following page)

FIG 3. (Continued). (E) Driver mutation (*y*-axis) frequencies (*x*-axis) for progressor (left side) and nonprogressor precursor samples (right side) in the training set. Mutations are color-coded by variant type, as shown in the figure legend. (F) Frequency of CN deletions (blue colors) and gains (red colors) for each cytoband (*y*-axis) for P-PC and NP-PC samples in the training data set. Significant differences are highlighted and markers with *. (G) The box plots, arranged from left to right, illustrate the differences between nonprogressor (green) and progressor (yellow) precursor samples for six metrics: GSS, total mutations, total number of structural variants focal deletions, focal gains, and APOBEC enrichment score. The Wilcoxon rank-sum test *P* values for the paired comparisons between the two groups are displayed at the top of each plot. CN, copy number; GSS, genomic scar score; MM, multiple myeloma; NDMM, newly diagnosed MM; NP-PC, nonprogressor plasma cell; P-PC, progressor plasma cell; SMM, smoldering multiple myeloma; SV, structural variants.

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We investigated whether events detected at only one time point were subclonal (excluding translocations). Of the 33 events (SNVs or CNAs), 20 (60%) were identified as subclonal changes at the time of detection, suggesting that they played no probable role in disease progression.

In comparing the overall changes between paired samples, we found that the total structural variant load was not significantly different between the paired samples (P = .31, paired Wilcoxon rank-sum test). However, the total SNV load (P = .016), GSS (P = .0008), focal deletions (P = .004), and APOBEC score (P = 3.1e-06) all showed a significant increase from SMM to MM (Fig 1B). The increases in GSS and focal deletions were primarily driven by acquired genomic deletions. Importantly, in those patients with short TTP (<median TTP) to MM, we did not observe significant differences in GSS and mutational load between the two groups, suggesting that short TTP samples diagnosed as SMM already has MM genome (Fig 1C, Data Supplement, Fig S1).

To evaluate the intratumor heterogeneity, we calculated and compared the area under the curves for cumulative mutation accumulation over CCFs. Reconstructing clonal evolution using paired samples revealed patterns of static and positive clonal evolution (Fig 2A). A paired sample comparison of CCF distributions revealed a slight but significant increase in clonal mutation frequencies at MM (P = .041, Data Supplement, Fig S1D) compared with the SMM from the same patients. Static evolution, in which clonal composition remains the same from SMM to MM, tended to be more common in patients with a short TTP (<1.52 years), but this was not statistically significant. Positive selection, in which a subclone shows positive selection, were also at a similar frequency between short and long (≥1.52 years, median time in all progressed samples) TTP (Figs 2B-2D). Among all 51 paired samples, 34 (66.6%) had the same clonal content and eight (15%) had enrichment of the existing clone detected in SMM. However, the accumulation of CCF between pairs is nearly identical (Fig 2B). This suggests that the majority of progressed SMMs were genomically similar to MM at diagnosis.

Comparison Between SMM and NDMM

We compared all available samples with genomic data in SMM and in NDMM to understand the differences between conditions. First, we used SNP arrays to examine CNA between 160 SMM and 336 NDMM patients at the cytoband level. This analysis revealed that the CN gains were similar in both groups (Fig 3A and Data Supplement, Fig S2). However, there were higher CN deletions observed in the genomes transitioned from SMM to MM (Figs 3B and 3C).

Overall, the number of translocations, excluding MYC, was also comparable between the two groups, suggesting that these events occur early in the course of plasma cell conditions. In terms of mutations, *NRAS* mutations were observed more frequently in MM compared with SMM (10% *v* 19%). Other notable driver mutations with $P \leq .1$ included *TP53* (1% in SMM, 7% in MM), *TENT5C* (*FAM46C*; 5% in SMM, 9% in MM), and *ATM* (2% in SMM, 4% in MM). *ROBO1* was the only driver mutation that exhibited a significantly higher frequency in SMM (Fig 3D).

Genomic Changes Between Nonprogressor and Progressor Patients

Differences in mutations most frequently observed in MM were also identified between P-PC and NP-PC. Most notably, fewer mutations in *KRAS* and *NRAS* and no mutations in *CYLD* and *FGFR3* were reported in NP-PC (Fig 3E). From SMM samples, 11 CN gain/deletion hotspots were identified using GISTIC (Data Supplement, Fig S3A). We observed that the frequency of del8p23.1 (P = .009) and gain1q21.1 (P = .06) was significantly lower in NP-PC compared with that of P-PC samples (Fig 3F, Data Supplement, Fig S3B). We found 23% of patients with *GATA4* deletions in P-PC versus 7.6% in NP-PC with (OR, 3.5 [95% CI, 1.3 to 9.8]; P = .01). This is similar to chr8p23.1 CN changes. Investigating the role of translocations among NP-PC versus P-PC and NDMM, we observed similar frequency of t(11;14) (18%, 10%, and 18%, respectively, P = .48 for P-PC *v* NP-PC, P = .99 for P-PC *v* NDMM), t(4;14) (21%, 10%, and 10%, respectively, P = .31 for P-PC *v* NP-PC, P = .09 for P-PC *v* NDMM), and t(14;16) (0%, 3%, and 2% respectively, P = .46 for P-PC *v* NP-PC, P = .99 for P-PC *v* NDMM) across samples. However, 29% of P-PC and 33% of NDMM compared with 6% of NP-PC samples had *MYC* translocation (P = .029 for P-PC *v* NP-PC, P = .69 for P-PC *v* NDMM); however, when *MYC* translocations were only restricted to immunoglobulin regions, there were no significant differences between the two groups (Data Supplement, Fig S3C).

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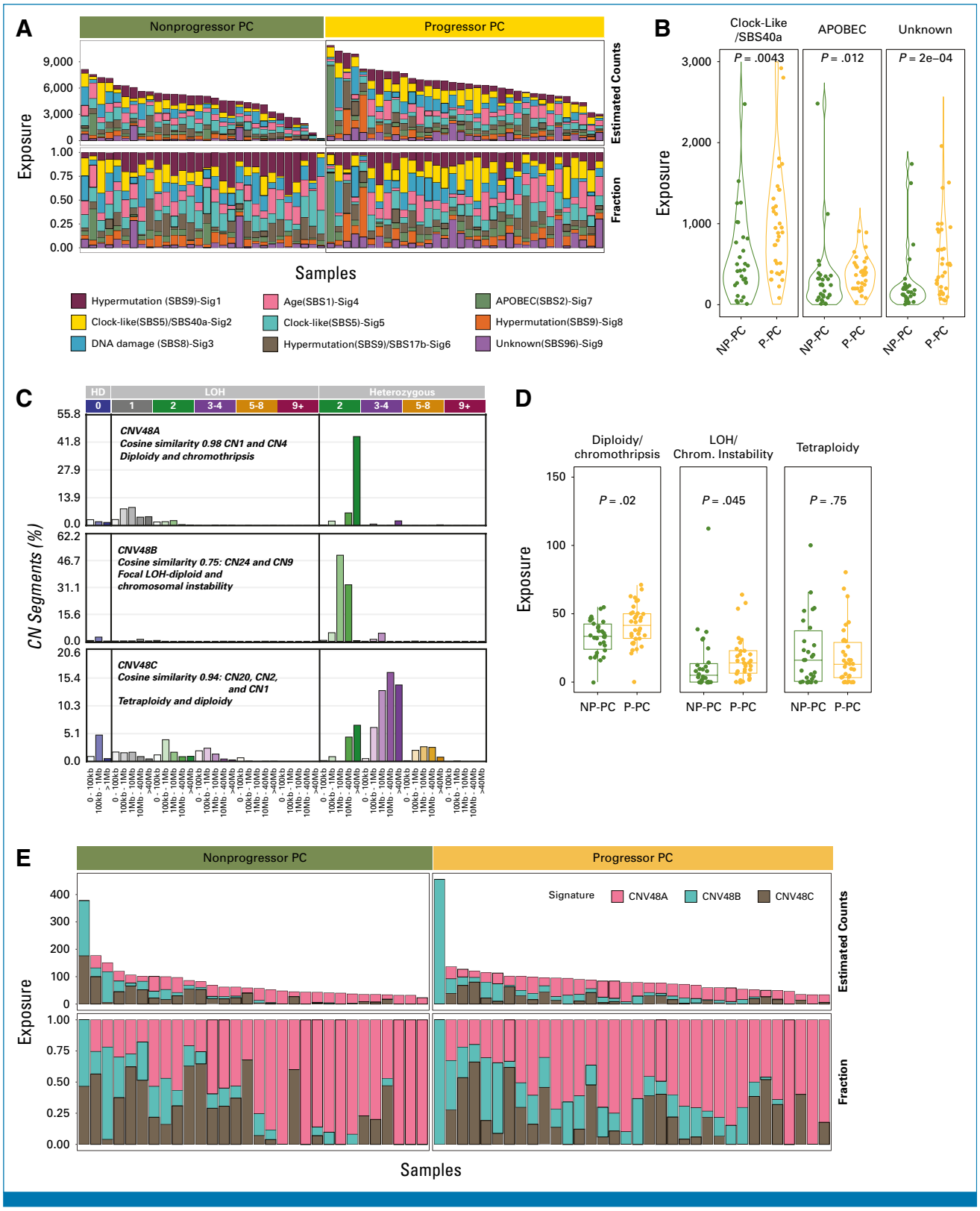

FIG 4. Mutational signature differences between nonprogressor and progressor precursor samples. (A) Absolute (upper panel) and relative (bottom panel) exposures of each mutation signature (color-coded as indicated in the figure legend) for nonprogressor (left) and progressor (right) precursor samples. The samples are arranged in decreasing order based on total mutational counts. (B) Estimated counts of absolute mutations (*y*-axis) are compared between nonprogressors (green) and progressors (yellow) across various mutational processes, displayed in each panel. The mutational processes and their corresponding *P*-values for comparison are indicated at the top of each panel. (C) Decomposition plots illustrate three CN signatures (CNV48A-C). The heterozygosity status and total CN (0 to 9+) are displayed above each column. Segment sizes are indicated at the bottom, whereas the frequency of (continued on following page)

FIG 4. (Continued). each event type is represented on the *y*-axis. (D) Estimated counts of absolute exposures of CNA signatures (*y*-axis) are compared between nonprogressors (green) and progressors (yellow) across three CNA processes, as shown in each panel. The mutational processes and their corresponding *P* values for comparison are indicated at the top of each panel. (E) Absolute (upper panel) and relative (bottom panel) exposures of each CNA signature (color-coded as indicated in the figure legend) for nonprogressor (left) and progressor (right) precursor samples. The samples are arranged in decreasing order based on total mutational counts. CN, copy number; CNA, copy number alterations; HD, homozygos deletion; LOH, loss of heterozygosity; NP-PC, nonprogressor plasma cell; P-PC, progressor plasma cell.

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We observed that P-PC had significantly higher GSS (paired sample Wilcoxon rank-sum test $P = .041$), total mutations and SVs ($P = .0002$ and .04), focal deletions ($P = .003$) and APOBEC enrichment ($P = .081$) than NP, whereas total genomic gains were not significantly different between the two groups ($P = .64$; Fig 3G). Consistently, GSS is further increased in short TTP (Data Supplement, Fig S1E).

Mutational and CNA Signatures Between P-PC and NP-PC

Using the SNVs, we asked whether mutational signature utilization differed between progressor and nonprogressor SMMs. Nine mutational signatures were identified in our samples (Fig 4A), with three being significantly different between the two groups, clock-like ($P = .004$) and APOBEC ($P = .012$), and an unspecified process had a significantly lower exposure in NP-PC (Fig 4B, Data Supplement, Fig S4A).

By analyzing SVs with the same framework, we identified 10 SV signatures (Data Supplement, Fig S4B). Three comprised nonclustered events, and seven included both clustered and nonclustered events. SV signatures 2 and 3, related to double-stranded breaks, were significantly lower in NP-PC (Data Supplement, Figs S4B-S4D). We investigated CN signatures and observed three CN signatures; signatures associated with chromothripsis (CNV48A) and chromosomal instability (CNV48B) had a significant increase in P-PC versus NP-PC, whereas the signature associated with hyperdiploidy (CNV48C) did not show significant differences (Figs 4C-4E). Both significant differences may be related to DNA-repair processes, whereas large segment gains are potentially early and related to mis-segregation of chromosomes. These CNV signatures are different from focal deletions or GSS; however, similar factors may contribute to them.

Genomic Model for the Prediction of Low-Risk SMM

We created a scoring system using five significantly different factors between P-PC and NP-PC (Figs 5A and 5B). The presence of these factors has shown significantly increased odds of developing MM based on comparison in our training data set (Fig 5A). The five genomic factors included in the scoring system are the presence of gain1q, CNA on chromosome 8, *NRAS* mutations, GSS > 5.5 (median in P-PC), and focal deletion >9.3 (median in P-PC). The score (the presence of each event is +1) is dichotomized in the training data set, and samples with score ≤1 are classified as low-risk SMM. This model achieved a stable accuracy, sensitivity, recall, and F1 scores confirmed by 1,000 bootstrapping and 70% subsampling in the training data set (Supplementary Methods). In an independent data set with 101 samples, our scoring system has achieved 85% sensitivity to classify NP-PC with a similar score in the training data set (Supplementary Methods). We tested the probability of progression between progressor and nonprogressors predicted by our model and found that there is a significant difference in progression probability in our validation cohort (likelihood ratio test $P = 1.4e-07$; Fig 5C). Next, we combined the genomic model with 20/2/20 for the patients with available information (Fig 5D). Combining the genomic model with 20/2/20 further increased the ability of both models. While 95% of patients predicted as high-risk SMM by both methods progressed within 4 years, only 24% of the patients predicted as low-risk by both methods progressed. Importantly, from 15 of 18 patients predicted as low-risk with 20/2/20 but high-risk by the genomic model progressed within 4 years (Fig 5E, likelihood ratio test $P = 4.42e-07$).

DISCUSSION

Diagnosis and management of precursor conditions are crucial in plasma cell disorders. Here, we report that most patients with SMM who progressed to MM showed only minor, subclonal genomic changes, indicating that MM alterations were already present in SMM. SMM patients with rapid progression had almost no new genomic changes, suggesting that the same dominant clones persisted. By contrast, nonprogressors over 5 years mostly had MGUS-like features.

Importantly, clonal selection was not required for progression in most SMM progressors; over 70% had the same clones at SMM and MM stages with minor differences. New driver gene mutations were rare, occurring in only a few patients, and it was subclonal. This suggests that deeper sequencing, when possible, may identify an even higher proportion of events being detected in longitudinal samples.

Comparing progressors and nonprogressors suggests monitoring specific markers over time, especially when a patient's risk level is uncertain. These progression markers involve

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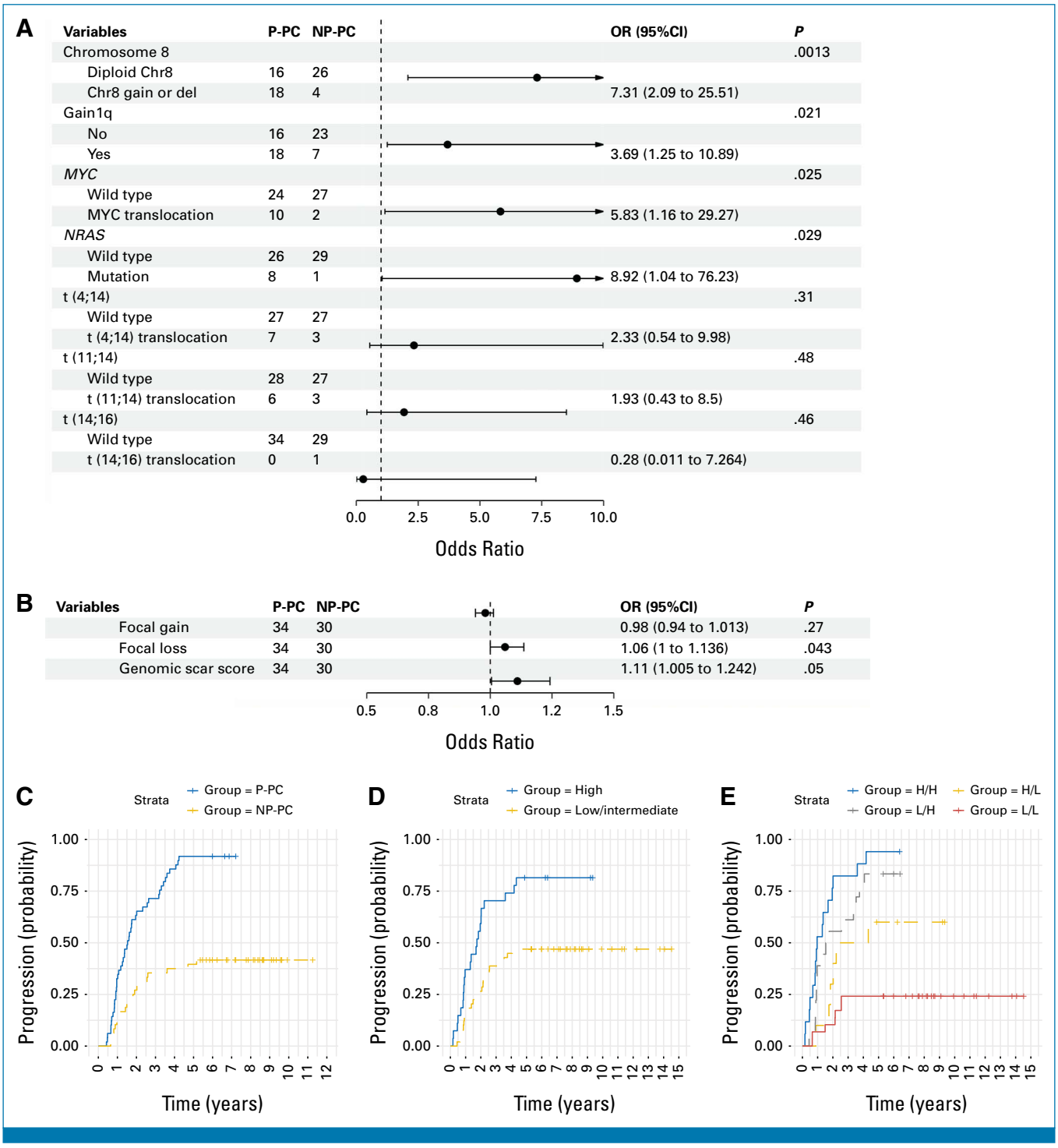

FIG 5. Prediction of low-risk SMM. (A) The forest plot illustrates the ORs (with 95% CIs) for genomic alterations and risk factors, comparing the low-risk and P-PC groups. The second and third columns display the counts of samples for each independent risk factor within the two groups: progressed PC and NP-PC. The fourth column presents the OR and their corresponding 95% CIs. Finally, the last column includes the *P* values obtained from Fisher's exact test. The vertical line represents an OR of 1. (B) The same as (A) but for continuous variables, GSS, focal deletions, and gains. Numbers in second and third columns represent the total number of samples in each group in the training data set. *P* values are calculated using the generalized linear model. (C) Probability of progression (*y*-axis) for P-PC and NP-PC predictions in the validation data set. The blue curve is for samples predicted as a progressor, and the yellow curve is for samples predicted as a nonprogressor. The *x*-axis shows the time since SMM diagnosis as years. (D) Probability of progression (*y*-axis) for high and low/intermediate groups defined by 20/2/20. The blue curve is for samples in the high-risk group, and the yellow curve is for samples in the low-/intermediate-risk group. The *x*-axis shows the time since SMM diagnosis as years. (E) Probability of progression (*y*-axis) for samples in four categories: (continued on following page)

FIG 5. (Continued). Blue indicates genomic model high and 20/2/20 high; gray indicates genomic model high and 20/2/20 low; yellow indicates genomic model low and 20/2/20 high; and red indicates low by both. The *x*-axis shows the time since SMM diagnosis as years. GSS, genomic scar score; NP-PC, nonprogressor plasma cell; OR, odds ratio; P-PC, progressor plasma cell; SMM, smoldering multiple myeloma.

increased genomic instability and nonrecurrent deletions. Surprisingly, the inter- and intratumor heterogeneity was similar in low- and high-risk groups. Notably, some MM risk events appear only at the subclonal level and do not directly drive SMM to MM progression but may predict survival after MM transformation. Compared with NDMM, SMM had higher *NRAS*, *TP53*, and *FAM46C* mutations.

Predicting the progression time from precursor stages is crucial task as early therapeutic interventions become available and effective risk stratification is needed to identify patients who require treatment versus those unlikely to progress, and hence, treatment would be unnecessary. Our findings indicate that genomic deletions are key indicators of disease progression. Most CN gains, except for gain1q, occur early, whereas deletions, except del13, are late events in myelomagenesis.[32,34] A significant increase in deletion events is observed as the disease approaches the symptomatic stage. This increased burden of deletions contributes to GSS. We have previously shown that low GSS and hyperdiploidy at diagnosis are associated with good-risk MM.[28] Similarly, lower GSS scores during precursor conditions are linked to longer or no progression, indicating that patients with high GSS are at risk of rapid advancement. CN signature analysis confirms similar observations for loss of heterozygosity and chromosomal instability. Our study, along with previous research, indicates a significant increase in APOBEC with disease progression and between high- and low-risk samples.

In our study, WGS offers higher resolution for significant genomic changes beyond driver events. It allows us to quantify mutational load, GSS, and increased deletions, which can be quantified with targeted panels or WES when modified for MM. This underscores the need for genome-wide analysis of precursor conditions. Previous studies by Bustoros et al,[11] Boyle et al,[15] and Bolli et al[14] have highlighted features like low *NRAS* mutation frequency and *MYC* translocations, which we also observed. However, WGS enabled us to identify additional genome-wide traits, which are linked to progression. Unlike previous reports, we did not observe changes in translocations or specific mutations, such as *TP53*, but this can change with a larger sample size.

MYC translocation frequency was between the progressor and nonprogressor groups with immunoglobulin genes; however, t(MYC) was increased when considering all translocations. While these translocations are typically seen as secondary events in transformed disease, both nonprogressors and progressors exhibited similar rates of primary translocations and CNAs, indicating that they are early events that require additional changes for conversion to myeloma. Interestingly, many genomic changes linked to poor outcomes in symptomatic MM were not statistically significant, indicating that they may not drive progression. However, this needs to be studied with larger data sets.

We have developed a model to identify patients with a low-risk SMM who are unlikely to progress to MM. This model uses five features: chr8 CNAs, *NRAS* mutation, gain1q, GSS, and focal deletions. It has shown high sensitivity and has been validated in an independent data set. We showed that combining the genomic model with the existing 20/2/20 clinical model increases the accuracy significantly, achieving high probability of progression or stable disease at 4 years when both models align. Discrepancies between the genomic and clinical models also shows added value of genomics in classifying patients with SMM. We acknowledge that one limitation of our study was the missing information because of long-term follow-ups, and the samples were selected from a tissue bank at a single center. Sample size in our discovery data set was also relatively small with multiple comparisons, which may cause potential false discoveries. Further investigations with larger sample size from multiple centers and systematic data collection can help addressing these challenges.

In summary, we describe important features highlighting genomic differences between progressor and nonprogressor precursor conditions using unique paired samples and define a good risk group with translational and clinical implications. Importantly, the lack of significant difference in mutations and similar CCF of mutation frequencies between paired samples indicates that plasma cells in high-risk patients already exhibit malignant changes. This challenges the current classification system based solely on clinical features and suggests a need to revise diagnostic criteria for SMM, particularly for those at high risk of progression. Given recent studies, similar to the AQUILA trial, emphasizing the importance of early treatment, an accurate risk stratification is now even more necessary. Our genomic model offers critical insights for improved risk stratification and can also identify patients unlikely to progress, thereby reducing unnecessary clinical interventions.

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Genomically Smoldering Multiple Myeloma Is Not a Distinct Entity But a Collection of Monoclonal Gammopathy of Undetermined Significance or Multiple Myeloma

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